

GGR276 Assignment 3

Part 1

Q1

The R^2 value (0.04014) shows that the relationship between population density and migration ratio is very weak. This means that the proportion of immigration can only explain about 4% of the population density change, which is extremely low, and there is almost no linear relationship between the two.

The following problems can be observed from the scatter chart: First, the data points are widely distributed and there is no obvious linear trend. Secondly, there are obvious anomalies in the data and most of the data points are concentrated in the low-value area, and a few extreme values lengthen the distribution range. Finally, the dispersion of the data indicates that there may be heterovariance.

It is not suitable for the linear regression model to fit the data set. The extremely low R^2 value and the lack of a clear linear pattern make the model very poor in explaining and predicting power. The existence of abnormal values may distort the regression results and reduce its reliability.

Q2

In this study, the dependent variable is population density, and the independent variable is median after-tax household income. Regression analysis should be performed because the goal of this study is to test whether median income can predict or explain population density. Regression analysis allows us to model the relationship between variables and quantify how changes in the independent variables affect the dependent variable. Correlation analysis measures the strength of the association between two variables without distinguishing between dependent and independent variables, while regression analysis explicitly models the dependent variable as a function of one or more independent variables. Furthermore, correlation analysis only indicates the strength and direction of the relationship, while regression analysis can be used to predict and explain changes in the dependent variable.

Q3

Null Hypothesis (H_0): There is no linear relationship between median after-tax household income and population density. In other words, the regression slope (β_1) equals 0.

Alternate Hypothesis (H_1): There is a linear relationship between median after-tax household income and population density. In other words, the regression slope (β_1) is not equal to 0.

Part 2

Q4

When arranging Median_a_2 in ascending order, the highlighted areas represent the census areas with the lowest median household income, indicating that the socio-economic status of these areas is relatively low. These areas are mainly concentrated around Brampton and parts of Mississauga. When arranging Population_Density in descending order, the highlighted area represents the area with the highest population density. These areas are mainly located in Brampton and Mississauga.

Some census areas appear in both rankings, and these areas with lower median income have high population density, indicating that there is a spatial overlap between low-income areas and densely populated urban areas.

Q5

Median after-tax household income:

H_0 (Null Hypothesis): Median household income is randomly distributed across the study area (no spatial autocorrelation).

H_1 (Alternate Hypothesis): Median household income is not randomly distributed and exhibits spatial autocorrelation (clustering).

Population density:

H_0 (Null Hypothesis): Population density is randomly distributed across the study area (no spatial autocorrelation).

H_1 (Alternate Hypothesis): Population density is not randomly distributed and exhibits spatial autocorrelation (clustering).

Q6

Median Income and Population Density in Peel Region



Q7

The central and southern areas of Peel are densely populated, especially near Brampton and Mississauga, and lower in the north and Caledon. There is a certain relationship between population density and the median family income. In some densely populated areas, such as central urban areas such as Brampton, the median family income is usually lower. In Caledon and other places, the relationship between population density and income is different.

Statistical methods such as correlation coefficient and regression analysis can be used to detect the direction and intensity of this correlation, and spatial self-correlation indicators such as the Moran index can be used to evaluate whether these patterns are spatially aggregated.

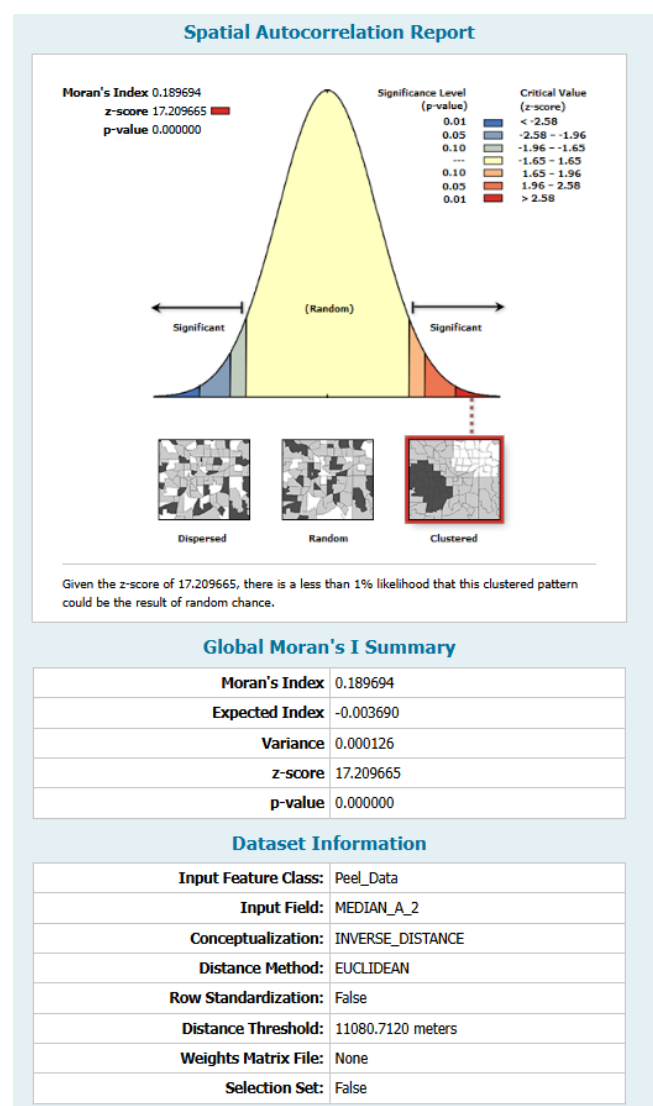
Part 3

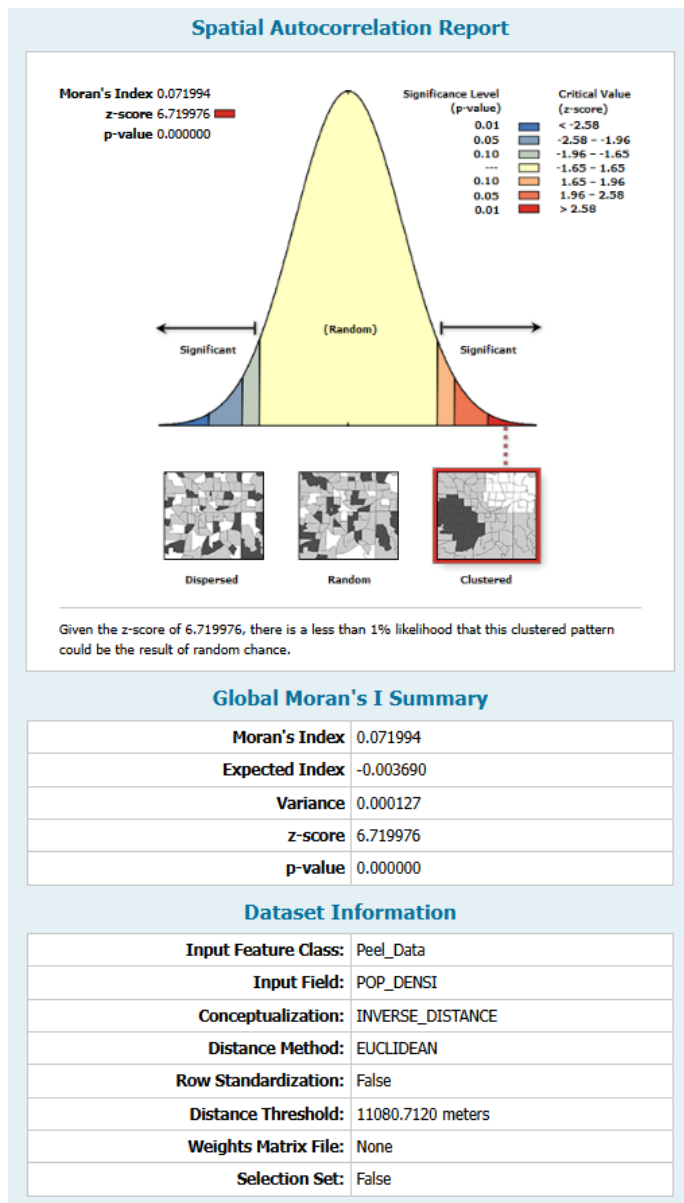
Q8

Spatial autocorrelation refers to the degree of correlation between spatial data values and adjacent positions. It measures whether similar values tend to be gathered together in space or randomly distributed. Positive spatial autocorrelation means that similar values are gathered in space, while negative spatial autocorrelation means that non-similar values are close to each other. Non-spatial correlation measures the relationship between two variables. It only evaluates whether two variables are statistically related without considering their geographical locations.

Spatial autocorrelation includes location and spatial proximity, and non-spatial correlation ignores spatial relationships and only focuses on attribute values.

Q9





For Median_a_2, the Moran's index is 0.189694, the z-score is 17.209665, and the p-value is 0.000000. A positive Moran's index indicates that similar values cluster together, meaning that high-income groups cluster together, and low-income groups also cluster together. The high z-score and zero p-value indicate statistical significance. For Population_Density, the Moran's index is 0.071994, the z-score is 6.719976, and the p-value is 0.000000. Although its Moran's index is lower than income's, it is still positive, indicating a spatial clustering pattern. The high z-score and zero p-value indicate that the observed pattern is statistically significant. High-density areas cluster together, and low-density areas are distributed across different regions. Overall, both variables show positive spatial autocorrelation, meaning that areas with similar values are close to each other.

This analysis tells us that the socioeconomic characteristics of Peel Region are not randomly distributed, but have a spatial structure. The government can use this pattern and structure to take targeted measures or policies for the region.

Part 4

Q10

Summary of OLS Results - Model Variables

Variable	Coefficient [a]	StdError	t-Statistic	Probability [b]	Robust_SE	Robust_t	Robust_Pr [b]
Intercept	111363.35482	2535.049219	43.929465	0.000000*	2781.655829	40.034915	0.000000*
POP_DENSI	-5.279470	1.081406	-4.882043	0.000003*	1.120913	-4.709975	0.000005*

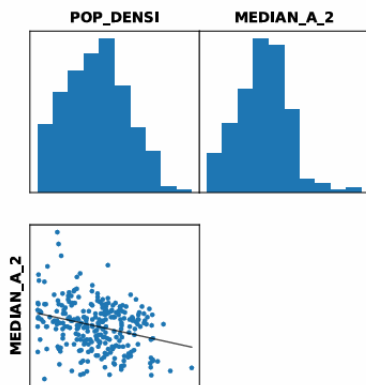
OLS Diagnostics

Input Features	Peel_Data	Dependent Variable	MEDIAN_A_2
Number of Observations	272	Akaike's Information Criterion (AICc)['d']	6146.485131
Multiple R-Squared['d']	0.081115	Adjusted R-Squared['d']	0.077712
Joint F-Statistic['e']	23.834345	Prob(>F), (1,270) degrees of freedom	0.000011*
Joint Wald Statistic['e']	22.183861	Prob(>chi-squared), (1) degrees of freedom	0.000002*
Koenker (BP) Statistic['f']	3.558717	Prob(>chi-squared), (1) degrees of freedom	0.059234
Jarque-Bera Statistic['g']	4.536669	Prob(>chi-squared), (2) degrees of freedom	0.103484

Notes on Interpretation

- * An asterisk next to a number indicates a statistically significant p-value ($p < 0.01$).
- [a] Coefficient: Represents the strength and type of relationship between each explanatory variable and the dependent variable.
- [b] Probability and Robust Probability (Robust_Pr): Asterisk (*) indicates a coefficient is statistically significant ($p < 0.01$); if the Koenker (BP) Statistic [f] is statistically significant, use the Robust Probability column (Robust_Pr) to determine coefficient significance.
- [c] Variance Inflation Factor (VIF): Large Variance Inflation Factor (VIF) values (> 7.5) indicate redundancy among explanatory variables.
- [d] R-Squared and Akaike's Information Criterion (AICc): Measures of model fit/performance.
- [e] Joint F and Wald Statistics: Asterisk (*) indicates overall model significance ($p < 0.01$); if the Koenker (BP) Statistic [f] is statistically significant, use the Wald Statistic to determine overall model significance.
- [f] Koenker (BP) Statistic: When this test is statistically significant ($p < 0.01$), the relationships modeled are not consistent (either due to non-stationarity or heteroskedasticity). You should rely on the Robust Probabilities (Robust_Pr) to determine coefficient significance and on the Wald Statistic to determine overall model significance.
- [g] Jarque-Bera Statistic: When this test is statistically significant ($p < 0.01$) model predictions are biased (the residuals are not normally distributed).

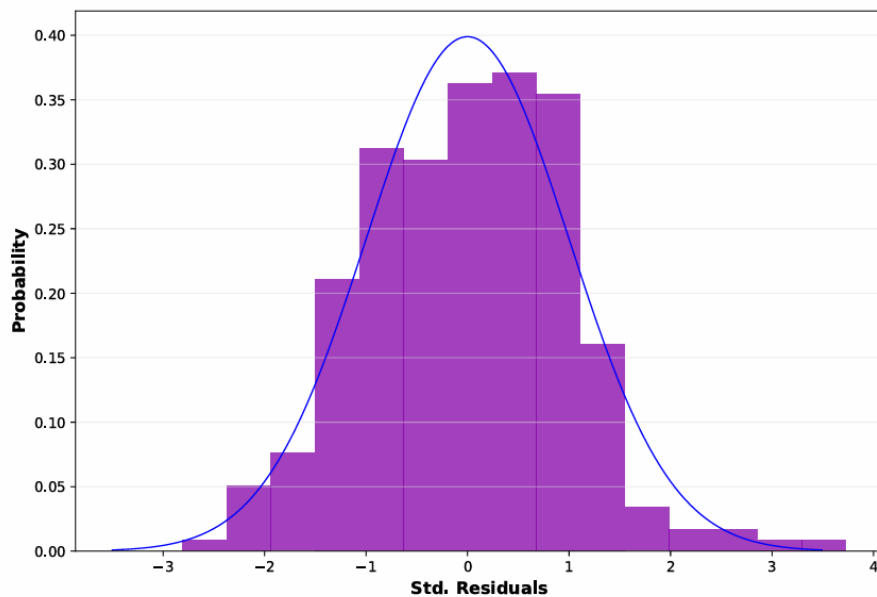
Variable Distributions and Relationships



The above graphs are Histograms and Scatterplots for each explanatory variable and the dependent variable. The histograms show how each variable is distributed. OLS does not require variables to be normally distributed. However, if you are having trouble finding a properly-specified model, you can try transforming strongly skewed variables to see if you get a better result.

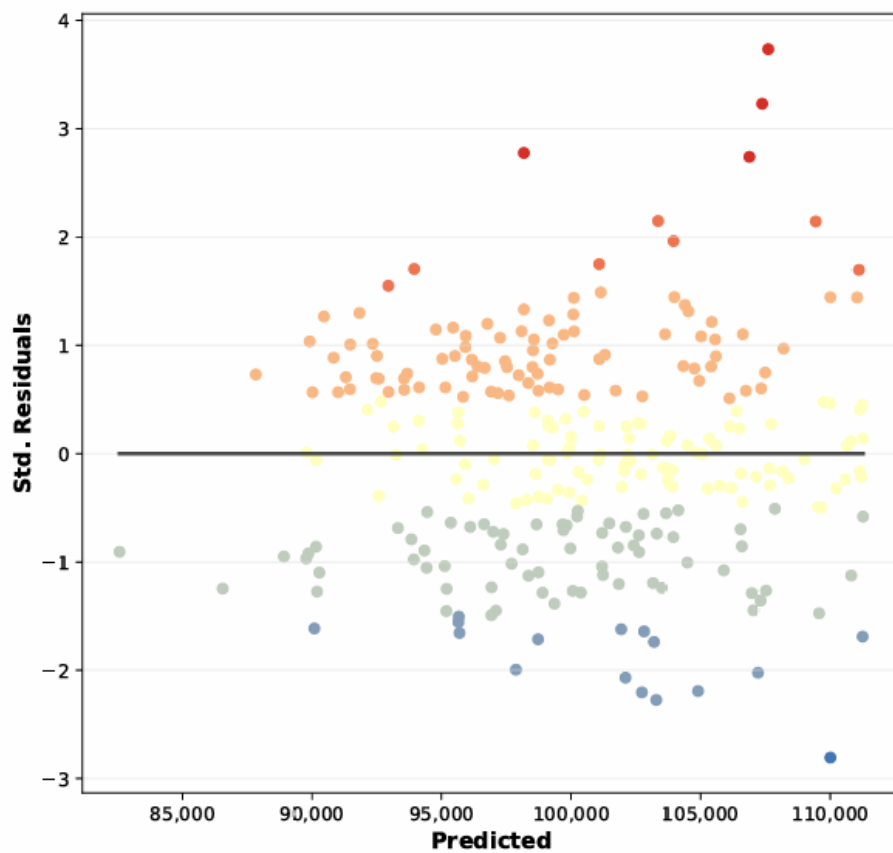
Each scatterplot depicts the relationship between an explanatory variable and the dependent variable. Strong relationships appear as diagonals and the direction of the slant indicates if the relationship is positive or negative. Try transforming your variables if you detect any non-linear relationships. For more information see the Regression Analysis Basics documentation.

Histogram of Standardized Residuals

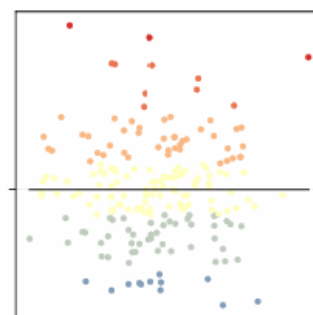


Ideally the histogram of your residuals would match the normal curve, indicated above in blue. If the histogram looks very different from the normal curve, you may have a biased model. If this bias is significant it will also be represented by a statistically significant Jarque-Bera p-value (*).

Residual vs. Predicted Plot



This is a graph of residuals (model over and under predictions) in relation to predicted dependent variable values. For a properly specified model, this scatterplot will have little structure, and look random (see graph on the right). If there is a structure to this plot, the type of structure may be a valuable clue to help you figure out what's going on.



Random Residuals

Ordinary Least Squares Parameters

Parameter Name	Input Value
Input Features	Peel_Data
Unique ID Field	FID2
Output Feature Class	C:\ArcGIS\GGR276_Assignment3\GGR276_Assignment3.gdb\Peel_Data_OrdinaryLeastSquares1
Dependent Variable	MEDIAN_A_2
Explanatory Variables	POP_DENSI
Selection Set	False

Q11

The regression model is: $\text{Median Income} = 111363.35 - 5.27947 \times \text{Population Density}$. This indicates a negative relationship between population density and median after-tax household income, meaning that as population density increases, median income tends to decrease. The Multiple R^2 value is 0.0811, which means that approximately 8.11% of the variation in median income can be explained by population density. This indicates that the model has low explanatory power. The Adjusted R^2 value 0.0777 is very similar, suggesting that the model is stable but still weak in explaining variation. The p-value for the slope is 0.000003, which is much smaller than 0.05. This indicates that the relationship between population density and median income is statistically significant. The p-value for the F-test is 0.000011, which shows that the overall regression model is statistically significant. The statistical results are significant, but the regression model is not robust because the R^2 value is very low. This suggests that other variables may also be influencing income patterns.

Q12

The null hypothesis is rejected for both median household income and population density.

For median income, the Moran's I value is 0.189694, with a z-score of 17.209665 and a p-value of 0.000000. For population density, the Moran's I value is 0.071994, with a z-score of 6.719976 and a p-value of 0.000000. The p-values are far below 0.05, meaning that the observed spatial patterns are statistically significant. In the maps produced in Part 2, high-density and lower-income areas concentrated in Brampton and Mississauga, and lower density in Caledon. Furthermore, the regression analysis also supports the presence of a statistically significant relationship between median income and population density (p-value for slope = 0.000003, p-value for F-test = 0.000011). Therefore, we reject the null hypothesis that these variables are randomly distributed and accept the alternative hypothesis that they exhibit spatial autocorrelation (clustering).